Monitoring Database Connections

Your Ruby on Rails application is a client application from the perspective of a PostgreSQL server instance. When you connect to it with psql, that's another client application.

SQL queries that are generated from Active Record code are sent to PostgreSQL using a database connection. Your application uses TCP over a network to communicate with the server. These are the basics, and a deep dive into networking out of our scope.

The total number of connections that can be served concurrently is limited based on the CPU and server resources of your PostgreSQL instance. Remember that PostgreSQL runs as a server process on a single server (computer) and it will use as much CPU, Memory, and Disk resources as it can based on how you've configured it.

In this section you'll take a high level look at how Active Record Ruby code is transformed into SQL statements and sent to the server.

1. You write Active Record Ruby code, chain methods together, there may be lazy evaluation, and eventually one or more SQL queries are produced as query text
2. The "driver" code ensures that the generated SQL query is PostgreSQL compatible
3. In Ruby on Rails the `pg` Ruby gem is a common adapter for connecting to PostgreSQL
4. Active Record has an application side Connection Pool, which is a pool of database connections. A connection is checked out from the pool in order to send a query to the server. The pool size expresses the maximum possible amount of connections that could be in use, although fewer could be in use at any given time.
5. Finally, the SQL queries are sent over the network using a connection. PostgreSQL computes a result, and in a normal scenario it's returned using the same connection. Other possibilities are that a statement does not complete successfully or is canceled.

Connections are opened on demand. The Active Record connection pool is a client and is responsible for closing the connection. If the connection is not closed it will be held open in an `idle` state indefinitely.

An `idle` connection can be in two states. It can be performing a transaction (a query happens inside an explicit or implicit transaction), or it can be idle and not performing a transaction.